

SOLIDWORKS API (.NET VBA C#) — Reference

(186/337): IModelDoc2 (part 9 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 9 of 17) (Type)

SOLIDWORKS API Help

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ICreatePlaneThruLineAndPt Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneThruLineAndPt Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneThruLineAndPt( _  
    ByVal AutoSize As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim AutoSize As System.Boolean  
Dim value As RefPlane
```

```
value = instance.ICreatePlaneThruLineAndPt(AutoSize)
```

C#

```
RefPlane ICreatePlaneThruLineAndPt(  
    System.bool AutoSize  
)
```

C++/CLI

```
RefPlane^ ICreatePlaneThruLineAndPt(  
& System.bool AutoSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AutoSize

True to automatically size the plane, false to not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneThruLineAndPt](#).

Remarks

This method uses the current document setting for displaying the reference plane. If display of reference planes is disabled, then you do not see the reference plane on the screen. If display of reference planes is enabled, then you see it. [IModelDocExtension::GetUserPreferenceToggle](#) swDisplayPlanes and [IModelDocExtension::SetUserPreferenceToggle](#) swDisplayPlanes get or set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns an IRefPlane object. This object can then be used for subsequent operations on the reference plane feature. Having a IRefPlane object may not be useful, except that it is a feature, which is an entity, so methods available on those objects are available. For an OLE user, those functions are directly accessible; for a COM user, those functions are available via QueryInterface. For example, to select the reference plane, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ViewDispRefPlanes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ICreatePlaneThruPtParallelToPlane Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneThruPtParallelToPlane Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneThruPtParallelToPlane( _  
    ByVal AutoSize As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim AutoSize As System.Boolean  
Dim value As RefPlane  
  
value = instance.ICreatePlaneThruPtParallelToPlane(AutoSize)
```

C#

```
RefPlane ICreatePlaneThruPtParallelToPlane(  
    System.bool AutoSize  
)
```

C++/CLI

```
RefPlane^ ICreatePlaneThruPtParallelToPlane(  
& System.bool AutoSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AutoSize

True to automatically size the plane, false to not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneThruPtParallelToPlane](#).

Remarks

This method uses the current document setting for displaying the reference plane. If display of reference planes is disabled, then you do not see the reference plane on the screen. If display of reference planes is enabled, then you see it. [IModelDocExtension::GetUserPreferenceToggle](#) swDisplayPlanes and [IModelDocExtension::SetUserPreferenceToggle](#) swDisplayPlanes get or set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns an IRefPlane object. This object can then be used for subsequent operations on the reference plane feature. Having a IRefPlane object may not be useful, except that it is a feature, which is an entity, so methods available on those objects are available. For an OLE user, those functions are directly accessible; for a COM user, those functions are available via QueryInterface. For example, to select the reference plane, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ViewDispRefplanes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ICreatePoint2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePoint2 Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreatePoint](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePoint2( _  
    ByVal PointX As System.Double, _  
    ByVal PointY As System.Double, _  
    ByVal PointZ As System.Double _  
) As SketchPoint
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim PointX As System.Double  
Dim PointY As System.Double  
Dim PointZ As System.Double  
Dim value As SketchPoint
```

```
value = instance.ICreatePoint2(PointX, PointY, PointZ)
```

C#

```
SketchPoint ICreatePoint2(  
    System.double PointX,  
    System.double PointY,  
    System.double PointZ  
)
```

C++/CLI

```
SketchPoint^ ICreatePoint2(  
& System.double PointX,  
& System.double PointY,  
& System.double PointZ  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointX
X location of the point
PointY
Y location of the point
PointZ
Z location of the point; ignored for 2D sketches

Return Value

Newly created [sketch point](#); this value is NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePoint2](#).

Remarks

This method creates a point in the active 2D or 3D sketch. If a sketch is not active, the point is not created and NULL is returned. Use [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::!GetActiveSketch2](#) to check to see if the sketch is active.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database. IModelDoc2::SetAddToDB also avoids inferencing.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePoint2 Method](#)
[ISketchPoint Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ICreateSpline Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateSpline Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::ICreateSpline](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSpline( _  
    ByVal PointCount As System.Integer, _  
    ByRef PointData As System.Double _  
    ) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim PointCount As System.Integer  
Dim PointData As System.Double  
Dim value As SketchSegment  
  
value = instance.ICreateSpline(PointCount, PointData)
```

C#

```
SketchSegment ICreateSpline(  
    System.int PointCount,  
    ref System.double PointData  
)
```

C++/CLI

```
SketchSegment^ ICreateSpline(  
& System.int PointCount,  
& System.double% PointData  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointCount

Number of points in the pointData array

PointData

Set of X,Y,Z point coordinates to use in creating the spline (see **Remarks**)

Return Value

Newly created [spline](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateSpline](#).

Remarks

This method creates a spline in the active 2D sketch. If a sketch is not active, then a new sketch is created. Use [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#) to check if the sketch active.

The PointData array is a set of, at least, two X, Y, Z values. The X value for the start point of the spline is PointData[0], the Y value for the start point is PointData[1], and the Z value for the start point is PointData[2]. The X value for the next point is PointData[3], and so on. For the COM interface, the total number of points in the array must be passed in. For the OLE interface, the total number of points are determined automatically, by taking the UBound of the PointData VARIANT and dividing by 3, so be careful to dimension that array correctly.

For COM applications, you can use the object pointer returned from this method to call any APIs on the [ISketchSegment](#) interface. You can obtain the underlying [ISketchSpline](#) object by using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchSpline object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call SketchSegment or [ISketchEllipse](#) APIs.

This method does not work with [IModelDoc2::SetAddToDB](#) or [IModelDoc2::SetDisplayWhenAdded](#). It always adds the spline directly to the database (as if IModelDoc2::SetAddToDB(True) was in effect), and you must redraw your document window to see the entities that you added (as if IModelDoc2::SetDisplayWhenAdded(false) was in effect).

In 2D sketches, SOLIDWORKS ignores the Z value in PointData.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateSplineByEqnParams Method](#)
[IModelDoc2::CreateSplinesByEqnParams Method](#)
[IModelDoc2::CreateSpline Method](#)
[IModelDoc2::ICreateSplineByEqnParams Method](#)
[IModelDoc2::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ICreateSplineByEqnParams Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateSplineByEqnParams Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::ICreateSplineByEqnParams](#).

.NET Syntax

```
Function ICreateSplineByEqnParams( _
    ByRef PropArray As System.Integer, _
    ByRef KnotsArray As System.Double, _
    ByRef CntrlPntCoordArray As System.Double _
) As SketchSegment
```



```
Dim instance As IModelDoc2
Dim PropArray As System.Integer
Dim KnotsArray As System.Double
Dim CntrlPntCoordArray As System.Double
Dim value As SketchSegment

value = instance.ICreateSplineByEqnParams(PropArray, KnotsArray, CntrlPntCoordArray)
```

```
SketchSegment ICreateSplineByEqnParams(
    ref System.int PropArray,
    ref System.double KnotsArray,
    ref System.double CntrlPntCoordArray
)
```

```
SketchSegment^ ICreateSplineByEqnParams(
    & System.int% PropArray,
    & System.double% KnotsArray,
    & System.double% CntrlPntCoordArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropArray
Includes dimension, order, number of control points, periodicity
KnotsArray
knot1, knot2, and so on
CntrlPntCoordArray
ontrolpoint1[dimension], controlpoint2[dimension], and so on

Return Value

Newly created [spline](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateSplineByEqnParams](#).

Remarks

The PropArray array contains four integers (placed in the first four doubles in a Visual Basic for Applications (VBA) SafeArray):

- Dimension
- Order
- Number of Control Points
- Periodicity (True or false)

The KnotsArray contains doubles with (Number of Control Points + Order) elements.

The size of the CntrlPntCoordArray array is based upon the curve dimension.

- Dimension = 2 then each Control Point is an array of 2 doubles (x, y)
- Dimension = 3 then each Control Point is an array of 3 doubles (x, y, z)
- Dimension = 4 then each Control Point is an array of 4 doubles (x, y, z, w) where w = weight

Pass the control point coordinates to this routine in sketch space. The Z value is interpreted as 0. In certain situations, you must transform your B-curve parameters to sketch space using [ISketch::ModelToSketchTransform](#).

NOTE: If the generated spline is a closed spline, then it must be flagged as periodic. Additionally, splines generated in sketches must be G1 continuous. If the data passed to this method does not generate a G1 continuous spline, then it is rejected and a spline is not created. If your data is not G1 continuous, then you must split the spline into multiple G1 segments and call this method for each segment.

You can use the returned object pointer directly to call any APIs on the [ISketchSegment](#) interface, or use QueryInterface to obtain the pointer to the ISketchSpline and any APIs on the [ISketchSpline](#) interface.

This method does not work with [IModelDoc2::SetAddToDB](#) or [IModelDoc2::SetDisplayWhenAdded](#). It always adds the spline directly to the database (as if IModelDoc2::SetAddToDB(True) is in effect), and you must redraw your document window to see the entities that you added (as IModelDoc2::SetDisplayWhenAdded(false) is in effect).

To create 3D splines, see [IModelDoc2::InsertCurveFilePoint](#) and related functions.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateSpline Method](#)
[IModelDoc2::CreateSplinesByEqnParams Method](#)
[IModelDoc2::ICreateSpline Method](#)
[IModelDoc2::ICreateSplinesByEqnParams Method](#)
[IModelDoc2::InsertCurveFile Method](#)
[IModelDoc2::InsertCurveFileBegin Method](#)
[IModelDoc2::InsertCurveFileEnd Method](#)
[IModelDoc2::ISketchSplineByEqnParams Method](#)
[IModelDoc2::SketchSplineByEqnParams2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ICreateSplinesByEqnParams Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateSplinesByEqnParams Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::ICreateSplinesByEqnParams](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSplinesByEqnParams( _  
    ByRef PropArray As System.Integer, _  
    ByRef KnotsArray As System.Double, _  
    ByRef CntrlPntCoordArray As System.Double _  
) As EnumSketchSegments
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim PropArray As System.Integer  
Dim KnotsArray As System.Double  
Dim CntrlPntCoordArray As System.Double  
Dim value As EnumSketchSegments  
  
value = instance.ICreateSplinesByEqnParams(PropArray, KnotsArray, CntrlPntCoordArray)
```

C#

```
EnumSketchSegments ICreateSplinesByEqnParams(  
    ref System.int PropArray,  
    ref System.double KnotsArray,  
    ref System.double CntrlPntCoordArray  
)
```

C++/CLI

```
EnumSketchSegments^ ICreateSplinesByEqnParams(  
& System.int% PropArray,  
& System.double% KnotsArray,  
& System.double% CntrlPntCoordArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropArray

Array of dimension, order, number of control points, periodicity (see **Remarks**)

KnotsArray

knot1, knot2, and so on (see **Remarks**)

CntrlPntCoordArray

controlpoint1[dimension], controlpoint2[dimension], and so on (see **Remarks**)

Return Value

Pointer to the newly created [splines enumeration](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateSplinesByEqnParams](#).

Remarks

The PropArray array contains four integers (placed in the first four doubles in a Visual Basic for Applications (VBA) SafeArray):

- Dimension
- Order
- Number of Control Points
- Periodicity (True or false)

The KnotsArray contains doubles with (Number of Control Points + Order) elements.

The size of the CntrlPntCoordArray array is based upon the curve dimension.

- Dimension = 2 then each Control Point is an array of 2 doubles (x, y)
- Dimension = 3 then each Control Point is an array of 3 doubles (x, y, z)
- Dimension = 4 then each Control Point is an array of 4 doubles (x, y, z, w) where w = weight

Pass the control point coordinates to this routine in sketch space. The Z value is interpreted as 0. In certain situations, you must transform your B-curve parameters to sketch space using [ISketch::ModelToSketchTransform](#).

NOTE: If the generated spline is a closed spline, then it must be flagged as periodic. Additionally, splines generated in sketches must be G1 continuous. If the data passed to this method does not generate a G1 continuous spline, then it is rejected and a spline is not created. If your data is not G1 continuous, then you must split the spline into multiple G1 segments and call this method for each segment.

You can use the returned object pointer directly to call any APIs on the [ISketchSegment](#) interface, or use QueryInterface to obtain the pointer to the ISketchSpline and any APIs on the [ISketchSpline](#) interface.

This method does not work with [IModelDoc2::SetAddToDB](#) or [IModelDoc2::SetDisplayWhenAdded](#). It always adds the spline directly to the database (as if IModelDoc2::SetAddToDB(True) is in effect), and you must redraw your document window to see the entities that you added (as IModelDoc2::SetDisplayWhenAdded(false) is in effect).

To create 3D splines, see [IModelDoc2::InsertCurveFilePoint](#) and related functions.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateSpline Method](#)
[IModelDoc2::CreateSplineByEqnParams Method](#)
[IModelDoc2::ICreateSpline Method](#)
[IModelDoc2::ICreateSplineByEqnParams Method](#)
[IModelDoc2::CreateSplinesByEqnParams Method](#)
[IModelDoc2::InsertCurveFile Method](#)
[IModelDoc2::InsertCurveFileBegin Method](#)
[IModelDoc2::InsertCurveFileEnd Method](#)
[IModelDoc2::SketchSplineByEqnParams2 Method](#)
[IModelDoc2::ISketchSplineByEqnParams Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IEditDimensionProperties3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IEditDimensionProperties3 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::IEditDimensionProperties](#).

Visual Basic (Declaration)

```
Function IEditDimensionProperties3( _
    ByVal TolType As System.Integer, _
    ByVal TolMax As System.Double, _
    ByVal TolMin As System.Double, _
    ByVal TolMaxFit As System.String, _
    ByVal TolMinFit As System.String, _
    ByVal UseDocPrec As System.Boolean, _
    ByVal Precision As System.Integer, _
    ByVal ArrowsIn As System.Integer, _
    ByVal UseDocArrows As System.Boolean, _
    ByVal Arrow1 As System.Integer, _
    ByVal Arrow2 As System.Integer, _
    ByVal PrefixText As System.String, _
    ByVal SuffixText As System.String, _
    ByVal ShowValue As System.Boolean, _
    ByVal CalloutText1 As System.String, _
    ByVal CalloutText2 As System.String, _
    ByVal CenterText As System.Boolean, _
    ByVal ConfigOption As System.Integer, _
    ByVal Count As System.Integer, _
    ByRef ConfigNames As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim TolType As System.Integer
Dim TolMax As System.Double
Dim TolMin As System.Double
Dim TolMaxFit As System.String
Dim TolMinFit As System.String
Dim UseDocPrec As System.Boolean
Dim Precision As System.Integer
Dim ArrowsIn As System.Integer
Dim UseDocArrows As System.Boolean
Dim Arrow1 As System.Integer
Dim Arrow2 As System.Integer
Dim PrefixText As System.String
Dim SuffixText As System.String
Dim ShowValue As System.Boolean
Dim CalloutText1 As System.String
Dim CalloutText2 As System.String
Dim CenterText As System.Boolean
Dim ConfigOption As System.Integer
Dim Count As System.Integer
Dim ConfigNames As System.String
Dim value As System.Boolean
```

```
value = instance.IEditDimensionProperties3(TolType, TolMax, TolMin, TolMaxFit, TolMinFit,
UseDocPrec, Precision, ArrowsIn, UseDocArrows, Arrow1, Arrow2, PrefixText, SuffixText, ShowValue,
CalloutText1, CalloutText2, CenterText, ConfigOption, Count, ConfigNames)
```

C#

```
System.bool IEditDimensionProperties3(
    System.int TolType,
    System.double TolMax,
    System.double TolMin,
    System.string TolMaxFit,
    System.string TolMinFit,
    System.bool UseDocPrec,
    System.int Precision,
    System.int ArrowsIn,
    System.bool UseDocArrows,
    System.int Arrow1,
```

```

        System.int Arrow2,
        System.string PrefixText,
        System.string SuffixText,
        System.bool ShowValue,
        System.string CalloutText1,
        System.string CalloutText2,
        System.bool CenterText,
        System.int ConfigOption,
        System.int Count,
        ref System.string ConfigNames
    )

```

C++/CLI

```

System.bool IEditDimensionProperties3(
& System.int TolType,
& System.double TolMax,
& System.double TolMin,
& System.String^ TolMaxFit,
& System.String^ TolMinFit,
& System.bool UseDocPrec,
& System.int Precision,
& System.int ArrowsIn,
& System.bool UseDocArrows,
& System.int Arrow1,
& System.int Arrow2,
& System.String^ PrefixText,
& System.String^ SuffixText,
& System.bool ShowValue,
& System.String^ CalloutText1,
& System.String^ CalloutText2,
& System.bool CenterText,
& System.int ConfigOption,
& System.int Count,
& System.String^% ConfigNames
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TolType

Type of tolerance as defined in [swTolType_e](#)

TolMax

Maximum value for the tolerance

TolMin

Minimum value for the tolerance

TolMaxFit

Text for the maximum FIT value when using a fit tolerance type

TolMinFit

Text for the minimum FIT value when using a fit tolerance type<

UseDocPrec

True to use the document's precision value, false to use the value specified for precision

Precision

Precision value to use for this dimension if UseDocPrec is false

ArrowsIn

Type of arrow to use on the first arrow of this dimension as defined in [swArrowStyle_e](#) if UseDocArrows is false

UseDocArrows

True to use the document's arrow types, false to use values specified for Arrow1 and Arrow2

Arrow1

Type of arrow to use on the first arrow of this dimension as defined in [swArrowStyle_e](#) if UseDocArrows is false

Arrow2

Type of arrow to use on the second arrow of this dimension as defined in [swArrowStyle_e](#) if UseDocArrows is false

PrefixText

Text for the prefix of the dimension

SuffixText

Text for the suffix of the dimension

ShowValue

True to display the value of the dimension in the user interface, false to not

CalloutText1

Callout text to display above the dimension

CalloutText2

Callout text to display below the dimension

CenterText

True if you want to center the text in the dimension, false if not

ConfigOption

Configuration options as defined [swInConfigurationOpts_e](#)

Count

Number of configurations

ConfigNames
Names of configurations to which to apply these edits (see **Remarks**)

Return Value

True if the dimension is modified, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::EditDimensionProperties3](#).

Remarks

This method always changes the dimension's parameters in the active configuration. For example, if you specify `swSpecifyConfiguration` for `ConfigOption` and do not specify the name of the active configuration in `configNames`, then the dimension's parameters in the active configuration are still affected.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::EditDimensionProperties3 Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

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IFeatureByPositionReverse Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IFeatureByPositionReverse Method (IModelDoc2)

Gets the *n*th from last feature in the document.

.NET Syntax

Visual Basic (Declaration)

```
Function IFeatureByPositionReverse( _  
    ByVal Num As System.Integer _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim Num As System.Integer  
Dim value As Feature
```

```
value = instance.IFeatureByPositionReverse(Num)
```

C#

```
Feature IFeatureByPositionReverse(  
    System.int Num  
)
```

C++/CLI

```
Feature^ IFeatureByPositionReverse(  
&    System.int Num  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Num

Number of feature from the last feature in the FeatureManager design tree; 0 is the last feature in FeatureManager design tree

Return Value

Pointer to the *n*th from last [feature](#) in the document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IFeatureByPositionReverse](#).

Remarks

This method returns features in the current order for the model, and this order changes when the model is edited.

Your application should not assume that:

- features retain the same relative or absolute position throughout the model's lifetime. For example, you should not assume that Sketch1 always appears before Sketch2.
- any feature has a specific name. Because features can be renamed, you cannot assume that the first reference plane feature is named Plane1.

When traversing the FeatureManager design tree, your application should use [IFeature::GetTypeName2](#) and [IFeature::GetSpecificFeature2](#) to identify specific features instead of relying solely on [IFeature::Name](#).

This method returns features in the model definition order, which is not the same as the order displayed in the user interface.

See [ITreeControlItem](#) for details.

Because SOLIDWORKS does not guarantee the name or positioning of default features, your application should not make any assumptions in this area. If your application is trying to access geometric features (i.e., sketches, fillets, bosses, reference surfaces, etc.) using [IModelDoc2::IFeatureByPositionReverse](#), then it is safest to determine the number of default features at the top and bottom of the list for each particular document. This could be done once for each document by traversing the FeatureManager design tree using [IModelDoc2::IFirstFeature](#) or [IModelDoc2::IFirstFeature](#) and [IFeature::GetNextFeature](#). Based on the feature type, [IFeature::GetTypeName](#), you can recognize where new features will be placed in the FeatureManager design tree upon creation.

For example, a new fillet is created at position (n-1) where n is the total number of features in the part. Therefore, to obtain this feature, then specify 1 for PositionFromEnd. This allows you to obtain the newly created fillet feature which is 1 from the bottom of the list.

If you are using this method to obtain the last feature object created by your application, then, as a precaution, you might also want to check the feature count immediately before your feature creation and immediately after your feature creation. If the feature count has increased by 1, then is relatively safe to assume that only your application has modified the document and added a feature. However, this is not a guaranteed methodology because another third-party applications might be running and might have also modified your document. Feature count can be determined by calling [IModelDoc2::GetFeatureCount](#). However, [IModelDoc2::GetFeatureCount](#) does not recognize Sheet-Metal and Flat-Pattern folders as features in the FeatureManager design tree. Sheet-Metal and Flat-Pattern folders were introduced in SOLIDWORKS 2013.

To access the first feature in the FeatureManager design tree and sub-features, use [IModelDoc2::IFirstFeature](#) and [IFeature::GetFirstSubFeature](#) methods, respectively.

[IModelDoc2::IFeatureByPositionReverse](#) can access suppressed features.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::FeatureByPositionReverse Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IFeatureFillet2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : [IFeatureFillet2 Method \(IModelDoc2\)](#)

Obsolete. Superseded by [IFeatureManager::FeatureCut](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IFeatureFillet2(  
    ByVal R1 As System.Double, _  
    ByVal Propagate As System.Boolean, _  
    ByVal Ftyp As System.Boolean, _  
    ByVal VarRadTyp As System.Boolean, _  
    ByVal OverflowType As System.Integer, _  
    ByVal NRadii As System.Integer, _  
    ByRef Radii As System.Double _  
) As System.Integer
```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim R1 As System.Double
Dim Propagate As System.Boolean
Dim Ftyp As System.Boolean
Dim VarRadTyp As System.Boolean
Dim OverflowType As System.Integer
Dim NRadii As System.Integer
Dim Radii As System.Double
Dim value As System.Integer

```

```
value = instance.IFeatureFillet2(R1, Propagate, Ftyp, VarRadTyp, OverflowType, NRadii, Radii)
```

C#

```

System.int IFeatureFillet2(
    System.double R1,
    System.bool Propagate,
    System.bool Ftyp,
    System.bool VarRadTyp,
    System.int OverflowType,
    System.int NRadii,
    ref System.double Radii
)

```

C++/CLI

```

System.int IFeatureFillet2(
    & System.double R1,
    & System.bool Propagate,
    & System.bool Ftyp,
    & System.bool VarRadTyp,
    & System.int OverflowType,
    & System.int NRadii,
    & System.double% Radii
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

R1

Propagate

Ftyp

VarRadTyp

OverflowType

NRadii

Radii

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IFeatureFillet2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IFeatureFillet3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IFeatureFillet3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::FeatureCut](#).

.NET Syntax

```

Function IFeatureFillet3( _
    ByVal R1 As System.Double, _
    ByVal Propagate As System.Boolean, _
    ByVal Ftyp As System.Integer, _
    ByVal VarRadTyp As System.Boolean, _
    ByVal OverflowType As System.Integer, _
    ByVal NRadii As System.Integer, _
    ByRef Radii As System.Double, _
    ByVal UseHelpPoint As System.Boolean, _
    ByVal UseTangentHoldLine As System.Boolean _
) As System.Integer

```



```

Dim instance As IModelDoc2
Dim R1 As System.Double
Dim Propagate As System.Boolean
Dim Ftyp As System.Integer
Dim VarRadTyp As System.Boolean
Dim OverflowType As System.Integer
Dim NRadii As System.Integer
Dim Radii As System.Double
Dim UseHelpPoint As System.Boolean
Dim UseTangentHoldLine As System.Boolean
Dim value As System.Integer

```

```

value = instance.IFeatureFillet3(R1, Propagate, Ftyp, VarRadTyp, OverflowType, NRadii, Radii,
UseHelpPoint, UseTangentHoldLine)

```

```

System.int IFeatureFillet3(
    System.double R1,
    System.bool Propagate,
    System.int Ftyp,
    System.bool VarRadTyp,
    System.int OverflowType,
    System.int NRadii,
    ref System.double Radii,
    System.bool UseHelpPoint,
    System.bool UseTangentHoldLine
)

```

```

System.int IFeatureFillet3(
    & System.double R1,
    & System.bool Propagate,
    & System.int Ftyp,
    & System.bool VarRadTyp,
    & System.int OverflowType,
    & System.int NRadii,
    & System.double% Radii,
    & System.bool UseHelpPoint,
    & System.bool UseTangentHoldLine
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

R1

Propagate

Ftyp

VarRadTyp

OverflowType

NRadii

Radii

UseHelpPoint

UseTangentHoldLine

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IFeatureFillet3](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IFeatureFillet4 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IFeatureFillet4 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::FeatureCut](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IFeatureFillet4( _  
    ByVal R1 As System.Double, _  
    ByVal Propagate As System.Boolean, _  
    ByVal UniformRadius As System.Boolean, _  
    ByVal Ftyp As System.Integer, _  
    ByVal VarRadTyp As System.Boolean, _  
    ByVal OverflowType As System.Integer, _  
    ByVal NRadii As System.Integer, _  
    ByRef Radii As System.Double, _  
    ByVal UseHelpPoint As System.Boolean, _  
    ByVal UseTangentHoldLine As System.Boolean, _  
    ByVal CornerType As System.Boolean, _  
    ByVal SetbackDistCount As System.Integer, _  
    ByRef SetBackDistances As System.Double _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim R1 As System.Double  
Dim Propagate As System.Boolean  
Dim UniformRadius As System.Boolean  
Dim Ftyp As System.Integer  
Dim VarRadTyp As System.Boolean  
Dim OverflowType As System.Integer  
Dim NRadii As System.Integer  
Dim Radii As System.Double  
Dim UseHelpPoint As System.Boolean  
Dim UseTangentHoldLine As System.Boolean  
Dim CornerType As System.Boolean  
Dim SetbackDistCount As System.Integer  
Dim SetBackDistances As System.Double  
Dim value As System.Integer
```

```
value = instance.IFeatureFillet4(R1, Propagate, UniformRadius, Ftyp, VarRadTyp, OverflowType,  
NRadii, Radii, UseHelpPoint, UseTangentHoldLine, CornerType, SetbackDistCount, SetBackDistances)
```

C#

```
System.int IFeatureFillet4(  
    System.double R1,  
    System.bool Propagate,
```

```

System.bool UniformRadius,
System.int Ftyp,
System.bool VarRadTyp,
System.int OverflowType,
System.int NRadii,
ref System.double Radii,
System.bool UseHelpPoint,
System.bool UseTangentHoldLine,
System.bool CornerType,
System.int SetbackDistCount,
ref System.double SetBackDistances
)

```

C++/CLI

```

System.int IFeatureFillet4(
& System.double R1,
& System.bool Propagate,
& System.bool UniformRadius,
& System.int Ftyp,
& System.bool VarRadTyp,
& System.int OverflowType,
& System.int NRadii,
& System.double% Radii,
& System.bool UseHelpPoint,
& System.bool UseTangentHoldLine,
& System.bool CornerType,
& System.int SetbackDistCount,
& System.double% SetBackDistances
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

R1

Propagate

UniformRadius

Ftyp

VarRadTyp

OverflowType

NRadii

Radii

UseHelpPoint

UseTangentHoldLine

CornerType

SetbackDistCount

SetBackDistances

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IFeatureFillet4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IFeatureFillet5 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IFeatureFillet5 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::FeatureCut](#).

Visual Basic (Declaration)

```
Function IFeatureFillet5( _
    ByVal Options As System.Integer, _
    ByVal R1 As System.Double, _
    ByVal Ftyp As System.Integer, _
    ByVal OverflowType As System.Integer, _
    ByVal NRadii As System.Integer, _
    ByRef Radii As System.Double, _
    ByVal SetbackDistCount As System.Integer, _
    ByRef SetBackDistances As System.Double, _
    ByVal PointCount As System.Integer, _
    ByRef PointRadiusArray As System.Double _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Options As System.Integer
Dim R1 As System.Double
Dim Ftyp As System.Integer
Dim OverflowType As System.Integer
Dim NRadii As System.Integer
Dim Radii As System.Double
Dim SetbackDistCount As System.Integer
Dim SetBackDistances As System.Double
Dim PointCount As System.Integer
Dim PointRadiusArray As System.Double
Dim value As System.Integer
```

```
value = instance.IFeatureFillet5(Options, R1, Ftyp, OverflowType, NRadii, Radii, SetbackDistCount,
SetBackDistances, PointCount, PointRadiusArray)
```

C#

```
System.int IFeatureFillet5(
    System.int Options,
    System.double R1,
    System.int Ftyp,
    System.int OverflowType,
    System.int NRadii,
    ref System.double Radii,
    System.int SetbackDistCount,
    ref System.double SetBackDistances,
    System.int PointCount,
    ref System.double PointRadiusArray
)
```

C++/CLI

```
System.int IFeatureFillet5(
    & System.int Options,
    & System.double R1,
    & System.int Ftyp,
    & System.int OverflowType,
    & System.int NRadii,
    & System.double% Radii,
    & System.int SetbackDistCount,
    & System.double% SetBackDistances,
    & System.int PointCount,
    & System.double% PointRadiusArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options

R1
Ftyp
v
OverflowType
v
NRadii
v
Radii
v
SetbackDistCount
v
SetBackDistances
v
PointCount
v
PointRadiusArray
v

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IFeatureFillet5](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IFeatureReferenceCurve Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IFeatureReferenceCurve Method (IModelDoc2)

Creates a reference curve feature from an array of curves.

.NET Syntax

Visual Basic (Declaration)

```
Function IFeatureReferenceCurve( _  
    ByVal NumOfCurves As System.Integer, _  
    ByRef BaseCurves As Curve, _  
    ByVal Merge As System.Boolean, _  
    ByVal FromFileName As System.String, _  
    ByRef ErrorCode As System.Integer _  
    ) As ReferenceCurve
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim NumOfCurves As System.Integer  
Dim BaseCurves As Curve  
Dim Merge As System.Boolean  
Dim FromFileName As System.String  
Dim ErrorCode As System.Integer  
Dim value As ReferenceCurve
```

```
value = instance.IFeatureReferenceCurve(NumOfCurves, BaseCurves, Merge, FromFileName, ErrorCode)
```

C#

```
ReferenceCurve IFeatureReferenceCurve(  
    System.int NumOfCurves,  
    ref Curve BaseCurves,  
    System.bool Merge,  
    System.string FromFileName,  
    out System.int ErrorCode  
)
```

C++/CLI

```
ReferenceCurve^ IFeatureReferenceCurve(  
& System.int NumOfCurves,  
& Curve^% BaseCurves,
```

```
& System.bool Merge,  
& System.String^ FromFileName,  
& [Out] System.int ErrorCode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfCurves

Number of curves from which to create the object

BaseCurves

Array of [curves](#)

Merge

True creates a single reference curve feature, false creates a reference curve feature for each curve in the array

FromFileName

Not used

ErrorCode

Error code as defined in [swFeatureError_e](#)

Return Value

[Reference curve](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IFeatureReferenceCurve](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::FeatureReferenceCurve Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IFirstFeature Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IFirstFeature Method (IModelDoc2)

Gets the first feature in the document.

.NET Syntax

Visual Basic (Declaration)

```
Function IFirstFeature() As Feature
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
Dim value As Feature
```

```
value = instance.IFirstFeature()
```

C#

```
Feature IFirstFeature()
```

C++/CLI

```
Feature^ IFirstFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [feature](#) in the document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IFirstFeature](#).

Example

[Access Assembly \(C++\)](#)
[Get Names of Creators of Features \(C++\)](#)

Remarks

This method:

- is identical to [IPartDoc::FirstFeature](#), but is available on the IModelDoc2 interface to include access to assemblies and drawings.
- can access the first feature, even if it is suppressed.
- returns features in the current order for the model, and this order changes when the model is edited.

Your application should not assume that:

- features retain the same relative or absolute position throughout the model's lifetime. For example, you should not assume that Sketch1 always appears before Sketch2.
- any feature has a specific name. Because features can be renamed, you cannot assume that the first reference plane feature is named Plane1.

When traversing the FeatureManager design tree, your application should use [IFeature::GetTypeNames2](#) and [IFeature::GetSpecificFeature2](#) to identify specific features instead of relying solely on [IFeature::Name](#).

This method returns features in the model definition order, which is not the same as the order displayed in the user interface. See [ITreeControlItem](#) for details.

To access the next feature in the FeatureManager design tree and subfeatures, use [IFeature::GetNextFeature](#) or [IFeature::!GetNextFeature](#) and [IFeature::GetFirstSubFeature](#) or [IFeature::!GetFirstSubFeature](#) methods, respectively.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetFeatureCount Method](#)
[IModelDoc2::FirstFeature Method](#)
[IModelDoc2::FeatureByPositionReverse Method](#)
[IModelDoc2::!FeatureByPositionReverse Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGet3rdPartyStorage Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGet3rdPartyStorage Method (IModelDoc2)

Gets the IStream interface to the specified third-party stream inside this SOLIDWORKS Design document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGet3rdPartyStorage( _  
    ByVal StringIn As System.String, _  
    ByVal IsStoring As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim StringIn As System.String  
Dim IsStoring As System.Boolean  
Dim value As System.Object
```

```
value = instance.IGet3rdPartyStorage(StringIn, IsStoring)
```

C#

```
System.Object IGet3rdPartyStorage(  
    System.string StringIn,  
    System.bool IsStoring  
)
```

C++/CLI

```
System.Object^ IGet3rdPartyStorage(
& System.String^ StringIn,
& System.bool IsStoring
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StringIn

Name of the stream; the name should be less than 30 characters and must be unique and qualified among all parties choosing to store within the current session

IsStoring

True if you are storing data, false if you are reading data

Return Value

IUnknown (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGet3rdPartyStorage](#).

Remarks

Call this interface when a SaveToStorageNotify or LoadFromStorageNotify notification is received. Do not call this method to save or load from storage in reaction to the FileSaveNotify or FileOpenNotify2 events.

During a file-open operation, your application receives a LoadFromStorageNotify event if you have registered for document notifications. When you receive this event, it is safe to open your Stream for reading using IModelDoc2::IGet3rdPartyStorage. When a file is fully loaded, you can call this method for reading your data at any time. This provides you with access to read your storage node whenever needed. However, it is not a good idea to attempt this while in the middle of a FileSaveNotify or FileOpenNotify2 as unforeseen conflicts can occur.

If your application is loaded in the middle of a SOLIDWORKS Design session, then you can traverse all open documents using ISldworks::EnumDocuments2, ISldWorks::IGetFirstDocument2, or IModelDoc2::IGetNext and read the storage from all existing open documents. From that point on, your application can use the LoadFromStorageNotify event to recognize when new documents have been opened and need reading.

For storage writing, IModelDoc2::IGet3rdPartyStorage interface is locked unless the SaveToStorageNotify event has been sent. Therefore, you can only write to your storage when the file is actually being saved and your application has received the SaveToStorageNotify event.

Your call to IModelDoc2::IGet3rdPartyStorage must be followed by a call to IModelDoc2::IRelease3rdPartyStorage. This must be done even when you fail to obtain a Stream and the return value from IModelDoc2::IGet3rdPartyStorage is NULL. If you fail to call IModelDoc2::IRelease3rdPartyStorage, the third-party node may remain locked and prevent future access. You are not required to call IModelDoc2::IRelease3rdPartyStorage, under any circumstance, if you called IModelDoc2::IGet3rdPartyStorage in reaction to one of the SaveToStorageNotify or LoadFromStorageNotify events. However, calling IModelDoc2::IRelease3rdPartyStorage excessively does not cause problems.

As work progresses during an active session and you have information that needs to be stored, you may want to flag the document as dirty using IModelDoc2::SetSaveFlag. This causes a **Save Changes?** prompt to display if the user tries to close the file without saving. If the user clicks **Yes** to this prompt, then your application receives the SaveToStorageNotify event.

NOTE: The name given to the storage stream should be registered so that no conflicts occur. Once registered, the stream name is reserved exclusively for your application.

Passing the unique ID string and a flag to determine if data is being stored or loaded returns an IUnknown pointer. You must then use QueryInterface() to get the IStream interface. The stream is used for serialization and then released in the third-party code.

```
SwRootStorage --|
|
|-- ThirdPty --|
|
|-- <SW Assigned IStream name 1>
|-- <SW Assigned IStream name 2>
|-- <SW Assigned IStream name 3>
...
|-- <SW Assigned IStream name n>
```

The IStream object used by the third party is written to IStream objects under an IStorage object called ThirdPty in the SOLIDWORKS compound document. Each third party writes to a single IStream object whose name is assigned by SOLIDWORKS.

NOTE: If you are using serialization, then be careful with the standard MFC macros; otherwise, you may get messages like **Unexpected File Format** after your application is unloaded. One way of using IMPLEMENT_SERIAL is:

```
IMPLEMENT_SERIAL( CCustomAttr, COBJ, VERSIONABLE_SCHEMA|0 )
```

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[DAssemblyDocEvents_LoadFromStorageNotify_Delegate](#)
[DAssemblyDocEvents_SaveToStorageNotifyEventHandler_Delegate](#)
[DDrawingDocEvents_LoadFromStorageNotifyEventHandler_Delegate](#)
[DDrawingDocEvents_SaveToStorageNotifyEventHandler_Delegate](#)
[DPartDocEvents_SaveToStorageNotifyEventHandler_Delegate](#)
[DPartDocEvents_LoadFromStorageNotifyEventHandler_Delegate](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetActiveConfiguration Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetActiveConfiguration Method (IModelDoc2)

Obsolete. Superseded by [IConfigurationManager.ActiveConfiguration](#).

.NET Syntax

Visual Basic (Declaration)

Function IGetActiveConfiguration() As Configuration

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As Configuration  
  
value = instance.IGetActiveConfiguration()
```

C#

Configuration IGetActiveConfiguration()

C++/CLI

Configuration^ IGetActiveConfiguration();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetActiveConfiguration](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetActiveSketch Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetActiveSketch Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::IGetActiveSketch2](#).

.NET Syntax

Visual Basic (Declaration)

Function IGetActiveSketch() As Sketch

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As Sketch
```

```
value = instance.IGetActiveSketch()
```

C#

```
Sketch IGetActiveSketch()
```

C++/CLI

```
Sketch^ IGetActiveSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetActiveSketch](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetActiveSketch2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetActiveSketch2 Method (IModelDoc2)

Gets the active sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetActiveSketch2() As Sketch
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As Sketch
```

```
value = instance.IGetActiveSketch2()
```

C#

```
Sketch IGetActiveSketch2()
```

C++/CLI

```
Sketch^ IGetActiveSketch2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Active [sketch](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetActiveSketch2](#).

Example

[Create Cylinder \(C++\)](#)

Remarks

Before you can use this method, you must select and activate a sketch. You can use [IModelDocExtension::SelectByID2](#) to select a sketch and [ISketchManager::InsertSketch](#) to make the sketch active.

For an example of getting a sketch through feature traversal, see [Get Sketches \(C++\)](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetActiveSketch2 Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAngularUnits Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetAngularUnits Method (IModelDoc2)

Gets the current angular unit settings.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAngularUnits() As System.Short
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Short
```

```
value = instance.IGetAngularUnits()
```

C#

```
System.short IGetAngularUnits()
```

C++/CLI

```
System.short IGetAngularUnits();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of 5 shorts (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The format of the returned data is an array of 5 shorts:

[*AngleUnit*, *FractionBase*, *FractionValue*, *SignificantDigits*, *RoundToFraction*]

where:

- *AngleUnit* = current angular units. You can find the unit types in [swAngleUnit.e](#).
- *FractionBase* = Not currently supported. Do not use the return value in this field.
- *FractionValue* = Not currently supported. Do not use the return value in this field.
- *SignificantDigits* = significant digits if using decimal units.
- *RoundToFraction* = Not currently supported. Do not use the return value in this field.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

[IModelDoc2::GetAngularUnits Method](#)
[IModelDoc2::!SetAngularUnits Method](#)
[IModelDoc2::SetAngularUnits Method](#)
[IModelDoc2::GetUnits Method](#)
[IModelDoc2::!GetUserUnit Method](#)
[IModelDoc2::!GetUnits Method](#)
[IModelDoc2::!GetUserUnit Method](#)
[IModelDoc2::SetUnits Method](#)
[IModelDoc2::LengthUnit Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetColorTable Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetColorTable Method (IModelDoc2)

Obsolete. Superseded by [ISldWorks::IGetColorTable](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetColorTable() As ColorTable
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As ColorTable
```

```
value = instance.IGetColorTable()
```

C#

```
ColorTable IGetColorTable()
```

C++/CLI

```
ColorTable^ IGetColorTable();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetColorTable](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetConfigurationByName Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetConfigurationByName Method (IModelDoc2)

Gets the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetConfigurationByName( _  
    ByVal Name As System.String _  
) As Configuration
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Name As System.String
Dim value As Configuration

value = instance.IGetConfigurationByName(Name)
```

C#

```
Configuration IGetConfigurationByName(
    System.string Name
)
```

C++/CLI

```
Configuration^ IGetConfigurationByName(
    & System.String^ Name
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name
Specified configuration name

Return Value

[Configuration](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetConfigurationByName](#).

Remarks

If the specified configuration has not been activated, then some data may be unavailable. For example, attempting to traverse assembly components for a configuration that has not been activated results in a NULL root component being returned from [IConfiguration::IGetRootComponent2](#). However, the IConfiguration object returned is useful for obtaining data that is stored with the IConfiguration object, such as the [IConfiguration::AlternateName](#) value. The specified configuration does not have to be activated to obtain this type of stored information.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddConfiguration3 Method](#)
[IModelDoc2::GetConfigurationCount Method](#)
[IModelDoc2::GetConfigurationNames Method](#)
[IModelDoc2::GetConfigurationByName Method](#)
[IModelDoc2::IGetConfigurationNames Method](#)
[IModelDoc2::DeleteConfiguration Method](#)
[IModelDoc2::ShowConfiguration2 Method](#)
[IModelDoc2::ConfigurationManager Property](#)
[IModelDoc2::IAddConfiguration3 Method](#)
[IConfiguration Interface](#)
[IConfigurationManager Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetConfigurationNames Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetConfigurationNames Method (IModelDoc2)

Gets the names of the configurations in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetConfigurationNames( _
    ByRef Count As System.Integer _
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Count As System.Integer
Dim value As System.String

value = instance.IGetConfigurationNames(Count)
```

C#

```
System.string IGetConfigurationNames(
    out System.int Count
)
```

C++/CLI

```
System.String^ IGetConfigurationNames(
    & [Out] System.int Count
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of configurations

Return Value

- in-process, unmanaged C++: Pointer to an array of the names of the configurations
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Use [IModelDoc2::GetConfigurationCount](#) to get the number of configurations in this part. Use the return value for Count to dimension the ConfigList array.

This method compares the number of configurations in the actual part with the number passed in. If the actual number of configurations is larger than the number passed in, no configurations are returned, and status returns S_ERROR. If the actual number of configurations is smaller than the number passed in, all of the configurations are returned in the configList, and configCount is changed to reflect the actual number of configurations.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddConfiguration3 Method](#)
[IModelDoc2::DeleteConfiguration2 Method](#)
[IModelDoc2::GetConfigurationByName Method](#)
[IModelDoc2::GetConfigurationNames Method](#)
[IModelDoc2::IAddConfiguration3 Method](#)
[IModelDoc2::IGetConfigurationByName Method](#)
[IModelDoc2::ShowConfiguration2 Method](#)
[IConfiguration Interface](#)
[IConfigurationManager Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetCoordinateSystemXformByName Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : [IGetCoordinateSystemXformByName Method \(IModelDoc2\)](#)

Obsolete. Superseded by [IModelDocExtension::GetCoordinateSystemTransformByName](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetCoordinateSystemXformByName( _  
    ByVal NameIn As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim NameIn As System.String  
Dim value As System.Double  
  
value = instance.IGetCoordinateSystemXformByName(NameIn)
```

C#

```
System.double IGetCoordinateSystemXformByName(  
    System.string NameIn  
)
```

C++/CLI

```
System.double IGetCoordinateSystemXformByName(  
& System.String^ NameIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NameIn

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetCoordinateSystemXformByName](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetCustomInfoNames Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetCustomInfoNames Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::CustomPropertyManager](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetCustomInfoNames() As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.String  
  
value = instance.IGetCustomInfoNames()
```

C#

```
System.string IGetCustomInfoNames()
```

C++/CLI

```
System.String^ IGetCustomInfoNames();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetCustomInfoNames](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetCustomInfoNames2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetCustomInfoNames2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::CustomPropertyManager](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetCustomInfoNames2( _  
    ByVal Configuration As System.String _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Configuration As System.String  
Dim value As System.String  
  
value = instance.IGetCustomInfoNames2(Configuration)
```

C#

```
System.string IGetCustomInfoNames2(  
    System.string Configuration  
)
```

C++/CLI

```
System.String^ IGetCustomInfoNames2(  
& System.String^ Configuration  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Configuration

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetCustomInfoNames2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDependencies Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetDependencies Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::GetDependencies2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDependencies( _  
    ByVal Traverseflag As System.Integer, _  
    ByVal Searchflag As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Traverseflag As System.Integer  
Dim Searchflag As System.Integer  
Dim value As System.String  
  
value = instance.IGetDependencies(Traverseflag, Searchflag)
```

C#

```
System.string IGetDependencies(  
    System.int Traverseflag,  
    System.int Searchflag  
)
```

C++/CLI

```
System.String^ IGetDependencies(  
& System.int Traverseflag,  
& System.int Searchflag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Traverseflag

Searchflag

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetDependencies](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDependencies2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetDependencies2 Method (IModelDoc2)

Gets all of the model's dependencies.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDependencies2( _  
    ByVal Traverseflag As System.Boolean, _  
    ByVal Searchflag As System.Boolean, _
```

```
ByVal AddReadOnlyInfo As System.Boolean _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Traverseflag As System.Boolean  
Dim Searchflag As System.Boolean  
Dim AddReadOnlyInfo As System.Boolean  
Dim value As System.String  
  
value = instance.IGetDependencies2(Traverseflag, Searchflag, AddReadOnlyInfo)
```

C#

```
System.string IGetDependencies2(  
    System.bool Traverseflag,  
    System.bool Searchflag,  
    System.bool AddReadOnlyInfo  
)
```

C++/CLI

```
System.String^ IGetDependencies2(  
& System.bool Traverseflag,  
& System.bool Searchflag,  
& System.bool AddReadOnlyInfo  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Traverseflag

True to traverse down into all dependent files, false for only the highest level within the dependencies

Searchflag

True to use the search rules to find dependencies, false to look where the documents were last saved

AddReadOnlyInfo

True to show read-only information in the returned strings, false to not

Return Value

- in-process, unmanaged C++: Pointer to an array of strings; there are two strings for each document returned in this list of dependent files:
 - first string is the file name
 - second string is the filename with the complete pathname

NOTE: This combination repeats itself for each dependent file found for this IModelDoc2.

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Example

[Get Model Dependencies \(C++\)](#)

Remarks

Calling this method on a drawing document returns a list of all the parts and assemblies used in the drawing. If you set Traverseflag to True, then each of the parts within the assembly file is also returned in this array of strings.

Because a derived mirror part is generated from another part, the list of model dependencies returned by this method would include the original part used to generate the derived mirror part.

Library features are completely unassociated. They do not require the library nor do they update when changes are made to the library feature. Therefore, this method does not return library features.

If you use this method with an assembly that contains two documents, Part1 and SubAssem1, an example of what might be returned is:

```
[ "Part1", "C:\temp\Part1.SLDPRT", "SubAssem1", "c:\temp\SubAssem1.SLDASM" ]
```

If Traverseflag is set to True, then all the documents contained in SubAssem1.ASM are returned. Also, suppressed components are recognized and returned by this method as a dependent file.

If AddReadOnlyInfo is set to True, then a string of True or false is added to the list of strings. If Part1 is read-only, then the resulting set of strings would be:

```
[ "Part1", "C:\\temp\\Part1.SLDPRT", "True", "SubAssem1", "c:\\temp\\SubAssem1.SLDASM", "" ]
```

NOTE: For assemblies containing suppressed or lightweight components, file references (return value from this method) point to the as-saved component reference because the actual component has not been loaded into memory. Because the suppressed and lightweight components have not actually been loaded, the current search criteria has not been applied to update the file references. Setting Searchflag to True vs. false causes this method to apply the current search criteria rules to locate the particular reference and may result in a different return value. Likewise unsuppressing or resolving these components causes the current search criteria to be applied and updates the assembly's component references in memory, if necessary.

When calling this method with Searchflag set to True, you can explicitly set the search folders first. This is important because this method uses the current SOLIDWORKS directory as its second search criteria. Because the user may have interactively opened files from some random directory, you cannot be certain the current SOLIDWORKS directory is pointing to the desired location. You may want to consider setting the search folders to that of the document being used in this method. For example:

```
// Get the current search folder settings so that you can reset them later

VARIANT_BOOL currentSearchFolderSetting = false;

BSTR currentSearchFolders = NULL;

hres = TheApplication->m_pSldWorks->GetUserPreferenceToggle ( swUseFolderSearchRules,
&currentSearchFolderSetting );

hres = TheApplication->m_pSldWorks->GetSearchFolders ( swDocumentType, &currentSearchFolders );


// Set the search folders to use the path of the document being queried with GetDependencies

BSTR bdocName;

hres = pModelDoc2->GetPathName(&bdocName);

CString docName(bdocName);


VARIANT_BOOL ok = false;

CString searchFolder = getFileLocation(docName); // Function to extract directory from full filename

hres = TheApplication->m_pSldWorks->SetSearchFolders ( swDocumentType, au2B(searchFolder), &ok );


// Reset search folders to previous values

hres = TheApplication->m_pSldWorks->SetUserPreferenceToggle ( swUseFolderSearchRules,
currentSearchFolderSetting );

hres = TheApplication->m_pSldWorks->SetSearchFolders ( swDocumentType, currentSearchFolders, &ok );
```

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetDependencies2 Method](#)
[IModelDoc2::GetNumDependencies Method](#)
[ISldWorks::GetDocumentDependencies2 Method](#)
[ISldWorks::IGetDocumentDependencies2 Method](#)
[ISldWorks::IGetDocumentDependenciesCount2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDesignTable Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetDesignTable Method (IModelDoc2)

Gets the design table associated with this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

Function IGetDesignTable() As DesignTable

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As DesignTable

value = instance.IGetDesignTable()
```

C#

```
DesignTable IGetDesignTable()
```

C++/CLI

```
DesignTable^ IGetDesignTable();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Design table](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetDesignTable](#).

Remarks

If you call this method from a drawing document, NULL is returned.

To access design tables from a drawing document you must get the [IModelDoc2](#) object associated with a particular drawing view and then call this method from that IModelDoc2 object. To determine if a drawing view has a design table associated with it, use [View::HasDesignTable](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CloseFamilyTable Method](#)
[IModelDoc2::DeleteDesignTable Method](#)
[IModelDoc2::GetDesignTable Method](#)
[IModelDoc2::InsertFamilyTableEdit Method](#)
[IModelDoc2::InsertFamilyTableNew Method](#)
[IModelDoc2::InsertFamilyTableOpen Method](#)
[IModelDoc2::IGetDesignTable Method](#)
[IModelDocExtension::HasDesignTable Method](#)
[IDesignTable Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDetailingDefaults Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetDetailingDefaults Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::GetUserPreferenceTextFormat](#) and [IModelDoc2::SetUserPreferenceTextFormat](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDetailingDefaults() As DetailingDefaults
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As DetailingDefaults

value = instance.IGetDetailingDefaults()
```

C#

```
DetailingDefaults IGetDetailingDefaults()
```

C++/CLI

```
DetailingDefaults^ IGetDetailingDefaults();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetDetailingDefaults](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetEntityName Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetEntityName Method (IModelDoc2)

Gets the name of the specified face, edge, or vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEntityName( _  
    ByVal Entity As Entity _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Entity As Entity  
Dim value As System.String
```

```
value = instance.IGetEntityName(Entity)
```

C#

```
System.string IGetEntityName(  
    Entity Entity  
)
```

C++/CLI

```
System.String^ IGetEntityName(  
& Entity^ Entity  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity
[Entity](#)

Return Value

Name of the entity

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetEntityName](#).

Example

[Get Entity Name \(C++\)](#)
[Get Named Entities \(C++\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetEntityName Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstAnnotation Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetFirstAnnotation Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::IGetFirstAnnotation2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstAnnotation() As Annotation
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As Annotation
```

```
value = instance.IGetFirstAnnotation()
```

C#

```
Annotation IGetFirstAnnotation()
```

C++/CLI

```
Annotation^ IGetFirstAnnotation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetFirstAnnotation](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstAnnotation2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetFirstAnnotation2 Method (IModelDoc2)

Gets the first annotation in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstAnnotation2() As Annotation
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As Annotation

value = instance.IGetFirstAnnotation2()
```

C#

```
Annotation IGetFirstAnnotation2()
```

C++/CLI

```
Annotation^ IGetFirstAnnotation2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [annotation](#) in model

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetFirstAnnotation2](#).

Remarks

For parts and assemblies, this method returns the first [IAnnotation](#) object in the model. For drawings, access the annotations using the [IView::GetFirstAnnotation3](#) method.

A dimension becomes suppressed or hidden when you specifically select a dimension and hide it, or when you select a feature and you select to hide all dimensions. If you need to filter out these dimensions, use [IAnnotation::Visible](#) to check that status.

NOTE: The difference between the obsoleted `IModelDoc2::GetFirstAnnotation` and this method, `IModelDoc2::GetFirstAnnotation2`, is that `IModelDoc2::GetFirstAnnotation2` retrieves any display dimension, including suppressed, hidden, or dangling dimensions.

To get the next annotation in the model, call [IAnnotation::GetNext3](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetFirstAnnotation2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstModelView Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetFirstModelView Method (IModelDoc2)

Gets the first view in a model document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstModelView() As ModelView
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As ModelView

value = instance.IGetFirstModelView()
```

C#

```
ModelView IGetFirstModelView()
```

C++/CLI

```
ModelView^ IGetFirstModelView();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [model view](#) in the document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetFirstModelView](#).

Remarks

You can traverse through the model views in a document by using this method to get the initial view and [IModelView::IGetNext](#) to get each of the following views. You have reached the end of the list when IModelView::GetNext returns NULL, .

See [IModelDoc2::EnumModelViews](#) for traversing the model views on a document.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelDoc2::GetModelViewCount Method](#)
[IModelDoc2::GetModelViewNames Method](#)
[IModelDoc2::IGetModelViewNames Method](#)
[IModelDoc2::ActiveView Property](#)
[IModelDoc2::IActiveView Property](#)
[IModelDoc2::ModelViewManager Property](#)
[IModelDoc2::EnumModelViews Method](#)
[IModelDocExtension::GetModelViewCount Method](#)
[IModelDocExtension::GetModelViews Method](#)
[IModelDocExtension::IGetModelViews Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetLayerManager Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetLayerManager Method (IModelDoc2)

Gets the layer manager ofr the current drawing document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLayerManager() As LayerMgr
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As LayerMgr
```

```
value = instance.IGetLayerManager()
```

C#

```
LayerMgr IGetLayerManager()
```

C++/CLI

```
LayerMgr^ IGetLayerManager();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Layer manager](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetLayerManager](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetLayerManager Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetLines Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetLines Method (IModelDoc2)

Gets all of the lines in the current sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLines() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Double  
  
value = instance.IGetLines()
```

C#

```
System.double IGetLines()
```

C++/CLI

```
System.double IGetLines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles:

$$\left[\text{LineType}, \text{StartPtX}, \text{StartPtY}, \text{StartPtZ}, \text{EndPtX}, \text{EndPtY}, \text{EndPtZ}, \dots \right]$$

where this array of 7 values repeats itself for each line in the current sketch. The number of doubles returned is (*lineCount* * 7). To determine the number of lines in the current sketch, use [IModelDoc2::GetLineCount](#).

See [swLineTypes_e](#) for valid line types.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetLineCount Method](#)
[IModelDoc2::GetLines Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMassProperties Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetMassProperties Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::IGetMassProperties](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMassProperties( _  
    ByRef MPropsData As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim MPropsData As System.Double  
Dim value As System.Boolean  
  
value = instance.IGetMassProperties(MPropsData)
```

C#

```
System.bool IGetMassProperties(  
    ref System.double MPropsData  
)
```

C++/CLI

```
System.bool IGetMassProperties(  
& System.double% MPropsData  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MPropsData

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetMassProperties](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMassProperties2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetMassProperties2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::IGetMassProperties](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMassProperties2( _  
    ByRef Status As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Status As System.Integer  
Dim value As System.Double  
  
value = instance.IGetMassProperties2(Status)
```

C#

```
System.double IGetMassProperties2(  
    out System.int Status  
)
```

C++/CLI

```
System.double IGetMassProperties2(  
    & [Out] System.int Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Status

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetMassProperties2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetModelViewNames Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetModelViewNames Method (IModelDoc2)

Gets a list containing the names of each model view in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetModelViewNames() As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.String  
  
value = instance.IGetModelViewNames()
```

C#

```
System.string IGetModelViewNames()
```

C++/CLI

```
System.String^ IGetModelViewNames();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array containing the names of each model view in this document
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetModelViewCount Method](#)
[IModelDoc2::GetModelViewNames Method](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelDoc2::IGetFirstModelView Method](#)
[IModelDoc2::ActiveView Property](#)
[IModelDoc2::IActiveView Property](#)
[IModelDoc2::ModelViewManager Property](#)
[IModelView Interface](#)
[IModelDoc2::EnumModelViews Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetNext Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetNext Method (IModelDoc2)

Gets the next document opened in the current SOLIDWORKS Design session.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNext() As ModelDoc2
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim value As ModelDoc2  
  
value = instance.IGetNext()
```

C#

```
ModelDoc2 IGetNext()
```

C++/CLI

```
ModelDoc2^ IGetNext();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Next [model document](#) opened in the current SOLIDWORKS Design session

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetNext](#).

Remarks

To traverse all documents open in this SOLIDWORKS Design session, the first IModelDoc2 object that calls this method must have been returned from [ISldWorks::IGetFirstDocument2](#).

NOTE: This method is available in datecode 1999/207 and later.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetNext Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetNumDependencies Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetNumDependencies Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::IGetNumDependencies2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNumDependencies( _  
    ByVal Traverseflag As System.Integer, _  
    ByVal Searchflag As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Traverseflag As System.Integer  
Dim Searchflag As System.Integer  
Dim value As System.Integer  
  
value = instance.IGetNumDependencies(Traverseflag, Searchflag)
```

C#

```
System.int IGetNumDependencies(  
    System.int Traverseflag,  
    System.int Searchflag  
)
```

C++/CLI

```
System.int IGetNumDependencies(  
& System.int Traverseflag,  
& System.int Searchflag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Traverseflag

Searchflag

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetNumDependencies](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetNumDependencies2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : [IGetNumDependencies2 Method \(IModelDoc2\)](#)

Gets the number of strings returned by [IModelDoc2::IGetDependencies2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNumDependencies2( _  
    ByVal Traverseflag As System.Boolean, _  
    ByVal Searchflag As System.Boolean, _  
    ByVal AddReadOnlyInfo As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Traverseflag As System.Boolean  
Dim Searchflag As System.Boolean  
Dim AddReadOnlyInfo As System.Boolean  
Dim value As System.Integer  
  
value = instance.IGetNumDependencies2(Traverseflag, Searchflag, AddReadOnlyInfo)
```

C#

```
System.int IGetNumDependencies2(  
    System.bool Traverseflag,  
    System.bool Searchflag,  
    System.bool AddReadOnlyInfo  
)
```

C++/CLI

```
System.int IGetNumDependencies2(  
& System.bool Traverseflag,  
& System.bool Searchflag,  
& System.bool AddReadOnlyInfo  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Traverseflag

True to traverse down into all dependent files, false for only the highest level within the dependencies

Searchflag

True to apply the current search criteria, false to return the dependent file information as it was stored

AddReadOnlyInfo

True to obtain read-only information with each dependent file

Return Value

Number of strings returned by [IModelDoc2::IGetDependencies2](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetNumDependencies2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetNumDependencies Method](#)
[IModelDoc2::GetDependencies2 Method](#)
[ISldWorks::GetDocumentDependencies2 Method](#)
[ISldWorks::IGetDocumentDependencies2 Method](#)
[ISldWorks::IGetDocumentDependenciesCount2 Method](#)

Availability

IGetRayIntersectionsPoints Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetRayIntersectionsPoints Method (IModelDoc2)

Gets the intersection point information generated by [IModelDoc2::IRayIntersections](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetRayIntersectionsPoints() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Double
```

```
value = instance.IGetRayIntersectionsPoints()
```

C#

```
System.double IGetRayIntersectionsPoints()
```

C++/CLI

```
System.double IGetRayIntersectionsPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array containing the hit points (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The array returned contains information about each hit of each ray on the given bodies, in the following format:

For each intersection found:

- (double) BodyIndex: index in BodiesIn array of the body that was hit
- (double) RayIndex: index in VectorsIn array of rays
- (double) IntersectionType: type of intersection found. See [swRayPtsResults_e](#) . Entry/Exit information (swRayPtsResultsENTER and swRayPtsResultsEXIT) are only returned when swRayPtsOptsENTRY_EXIT is specified in the options argument in the [IModelDoc2::IRayIntersections](#) call. If more than one intersection type is found, then the IntersectionType contains an OR'd set of value: from the swRayPtsResults_e enumeration.
- (double) x,y,z: coordinates of hit point
- (double) nx,ny,nz: normal (optionally appears when swRayPtsOptsNORMALS is specified in the Options argument in the IModelDoc2::RayIntersections call).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetRayIntersectionsPoints Method](#)
[IModelDoc2::GetRayIntersectionsTopology Method](#)
[IModelDoc2::IGetRayIntersectionsTopology Method](#)
[IModelDoc2::IMultiSelectByRay Method](#)
[IModelDoc2::ISelectByRay Method](#)
[IModelDoc2::MultiSelectByRay Method](#)
[IModelDoc2::SelectByRay Method](#)
[IModelDoc2::RayIntersections Method](#)

Availability

IGetRayIntersectionsTopology Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetRayIntersectionsTopology Method (IModelDoc2)

Gets the topology intersections generated by [IModelDoc2::IRayIntersections](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetRayIntersectionsTopology() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Object
```

```
value = instance.IGetRayIntersectionsTopology()
```

C#

```
System.object IGetRayIntersectionsTopology()
```

C++/CLI

```
System.Object^ IGetRayIntersectionsTopology();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array containing the hit entities (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This method only returns information if the swRayPtsOptsTOPOLS option was specified in the Options argument of [IModelDoc2::RayIntersections](#).

The array returned contains information about each entity hit of each ray on the given bodies. The topology hit can be coordinated with the hit point using the BodyIndex return value from [IModelDoc2::IGetRayIntersectionsPoints](#). You can determine the type of item hit (face, edge or vertex) by checking the IntersectionType return value in the IModelDoc2::IGetRayIntersectionsPoints.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetRayIntersectionsPoints Method](#)
[IModelDoc2::GetRayIntersectionsTopology Method](#)
[IModelDoc2::IGetRayIntersectionsPoints Method](#)
[IModelDoc2::MultiSelectByRay Method](#)
[IModelDoc2::ISelectByRay Method](#)
[IModelDoc2::MultiSelectByRay Method](#)
[IModelDoc2::SelectByRay Method](#)
[IModelDoc2::RayIntersections Method](#)
[IModelDoc2::IRayIntersections Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

IGetStandardViewRotation Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetStandardViewRotation Method (IModelDoc2)

Gets the specified view orientation matrix with respect to the Front view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetStandardViewRotation( _  
    ByVal ViewId As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ViewId As System.Integer  
Dim value As System.Double  
  
value = instance.IGetStandardViewRotation(ViewId)
```

C#

```
System.double IGetStandardViewRotation(  
    System.int ViewId  
)
```

C++/CLI

```
System.double IGetStandardViewRotation(  
    & System.int ViewId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewId

View ID that you want to evaluate as defined in [swStandardViews_e](#)

Return Value

Array describing the view rotation with respect to the Front view; this is an array of 9 doubles

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetStandardViewRotation](#).

Remarks

The end-user may have redefined the Front view to be something other than the X-Y plane.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetStandardViewRotation Method](#)
[IModelDocExtension::GetNamedViewRotation Method](#)
[IModelDocExtension::IGetNamedViewRotation Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetUnits Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetUnits Method (IModelDoc2)

Gets the current unit settings, fraction base, fraction value, and significant digits.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetUnits() As System.Short
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim value As System.Short

value = instance.IGetUnits()
```

C#

```
System.short IGetUnits()
```

C++/CLI

```
System.short IGetUnits();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of 5 shorts (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Example

[Get and Set Units \(C++\)](#)

Remarks

The format of the returned data is an array of shorts:

[*LengthUnit*, *FractionBase*, *FractionValue*, *SignificantDigits*, *RoundToFraction*]

where:

- *LengthUnit* = current model units as defined in [swLengthUnit_e](#).
- *FractionBase* = decimal or fraction as defined in [swFractionDisplay_e](#); fractional units are only valid if the model units are set to swINCHES or swFEETINCHES.
- *FractionValue* = denominator value if using fractional units.
- *SignificantDigits* = significant digits if using decimal units.
- *RoundToFraction* = flag denoting whether or not to round to the fraction; for example, if 4 was the denominator value and the actual value was .27, it is rounded to .25

You can also use [IModelDoc2::LengthUnit](#), which provides access to the LengthUnit parameter.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetAngularUnits Method](#)
[IModelDoc2::GetUnits Method](#)
[IModelDoc2::IGetAngularUnits Method](#)
[IModelDoc2::IGetUserUnit Method](#)
[IModelDoc2::GetUserUnit Method](#)
[IModelDoc2::ISetAngularUnits Method](#)
[IModelDoc2::SetAngularUnits Method](#)
[IModelDoc2::SetUnits Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetUserPreferenceTextFormat Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetUserPreferenceTextFormat Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetUserPreferenceTextFormat](#).

.NET Syntax

```
Function IGetUserPreferenceTextFormat(  
    ByVal UserPreferenceValue As System.Integer _  
) As TextFormat
```



```
Dim instance As IModelDoc2  
Dim UserPreferenceValue As System.Integer  
Dim value As TextFormat  
  
value = instance.IGetUserPreferenceTextFormat(UserPreferenceValue)
```

```
TextFormat IGetUserPreferenceTextFormat(  
    System.int UserPreferenceValue  
)
```

```
TextFormat^ IGetUserPreferenceTextFormat(  
    & System.int UserPreferenceValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreferenceValue

Value as defined in [swUserPreferenceTextFormat_e](#)

Return Value

[Text format](#) associated with UserPreferenceValue

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetUserPreferenceTextFormat](#).

Remarks

This method is equivalent to interactively getting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

IGetUserUnit Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetUserUnit Method (IModelDoc2)

Gets this document's units settings.

.NET Syntax

```
Function IGetUserUnit( _
    ByVal UnitType As System.Integer _
) As UserUnit
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim UnitType As System.Integer
Dim value As UserUnit

value = instance.IGetUserUnit(UnitType)
```

C#

```
UserUnit IGetUserUnit(
    System.int UnitType
)
```

C++/CLI

```
UserUnit^ IGetUserUnit(
    & System.int UnitType
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UnitType
Document unit type as defined in [swUserUnitsType_e](#)

Return Value

[IUserUnit](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetUserUnit](#).

Remarks

The properties of IUserUnit are read only. Set them using the IModelDocExtension::SetUserPreference* methods.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetAngularUnits Method](#)
[IModelDoc2::GetUnits Method](#)
[IModelDoc2::IGetUserUnit Method](#)
[IModelDoc2::IGetAngularUnits Method](#)
[IModelDoc2::IGetUnits Method](#)
[IModelDoc2::ISetAngularUnits Method](#)
[IModelDoc2::SetAngularUnits Method](#)
[IModelDoc2::SetUnits Method](#)
[ISldWorks::IGetUserUnit Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetVersionHistoryCount Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IGetVersionHistoryCount Method (IModelDoc2)

Gets the size of the array required to hold data returned by [IModelDoc2::IVersionHistory](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetVersionHistoryCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Integer

value = instance.IGetVersionHistoryCount()
```

C#

```
System.int IGetVersionHistoryCount()
```

C++/CLI

```
System.int IGetVersionHistoryCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Size of array required for the version history

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IGetVersionHistoryCount](#).

Example

[Document Version History \(C++\)](#)

Remarks

If the document has not yet been saved, then no version history information exists and this method returns 0.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::VersionHistory Method](#)
[ISldWorks::VersionHistory Method](#)
[ISldWorks::VersionHistory Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBOMBalloon2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : [InsertBOMBalloon2 Method \(IModelDoc2\)](#)

Obsolete. Superseded by [IModelDocExtension::InsertBOMBalloon](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertBOMBalloon2( _
    ByVal Style As System.Integer, _
    ByVal Size As System.Integer, _
    ByVal UpperTextStyle As System.Integer, _
    ByVal UpperText As System.String, _
    ByVal LowerTextStyle As System.Integer, _
    ByVal LowerText As System.String _
) As Note
```

```

Dim instance As IModelDoc2
Dim Style As System.Integer
Dim Size As System.Integer
Dim UpperTextStyle As System.Integer
Dim UpperText As System.String
Dim LowerTextStyle As System.Integer
Dim LowerText As System.String
Dim value As Note

value = instance.IInsertBOMBalloon2(Style, Size, UpperTextStyle, UpperText, LowerTextStyle, LowerText)

```

C#

```

Note IInsertBOMBalloon2(
    System.int Style,
    System.int Size,
    System.int UpperTextStyle,
    System.string UpperText,
    System.int LowerTextStyle,
    System.string LowerText
)

```

C++/CLI

```

Note^ IInsertBOMBalloon2(
    & System.int Style,
    & System.int Size,
    & System.int UpperTextStyle,
    & System.String^ UpperText,
    & System.int LowerTextStyle,
    & System.String^ LowerText
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Style

Balloon style as defined in [swBalloonStyle_e](#)

Size

Balloon size as defined in [swBalloonFit_e](#)

UpperTextStyle

Text style for the upper text of the balloon as defined in [swBalloonTextContent_e](#)

UpperText

Text string to be placed in the upper text of the balloon

LowerTextStyle

Text style for the lower text of the balloon as defined in [swBalloonTextContent_e](#)

LowerText

Text string to be placed in the lower text of the balloon

Return Value

Newly created [note](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IInsertBOMBalloon2](#).

Remarks

The selected object can be an edge, silhouette, face, or vertex.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IInsertBOMBalloon2 Method](#)

